Open-Source Development

ITIC2262

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4. Analyze Features of Open-Source Products

* Identifying and Using Features in Open-Source Development

Intended Learning Outcomes

* After this lesson students will be able to Working with the Open-source products and Analyze features of the open-source products :
  - Explain Open Source Software (OSS)
  - Identify features of OSS products
  - Analyze advantages and disadvantages
  - Compare OSS with proprietary software
  - Understand OSS development practices
  - Evaluate suitable OSS products for different applications

What are Open-Source Software

* **Definition**

Open Source Software (OSS) is software whose source code is publicly available and can be:

- Viewed
- Modified
- Improved
- Distributed







* **Eg:**
  - Linux
  - Mozilla Firefox
  - LibreOffice
  - VLC Media Player
  - WordPress









Characteristics of Open Source Software

* **Main characteristics**

An Operating System (OS) manages computer hardware and software resources while providing services for applications.

- Community-driven
- Transparent
- Flexible
- Secure
- Collaborative development
- Free redistribution

Source Code

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Anyone Can Review

↓

Modify

↓

Improve

↓

Share

Major Features of Open Source Products

Feature	Description
Free access	No license fee
Source code availability	Developers can inspect code
Community support	Global collaboration
Security	Vulnerabilities found quickly
Customization	Adapt to user requirements
Interoperability	Works with many systems
Scalability	Suitable from small to enterprise use

Open Source Development Mode

- Development Cycle :

Developer

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Upload Code

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Git Repository

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Community Review

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Bug Reports

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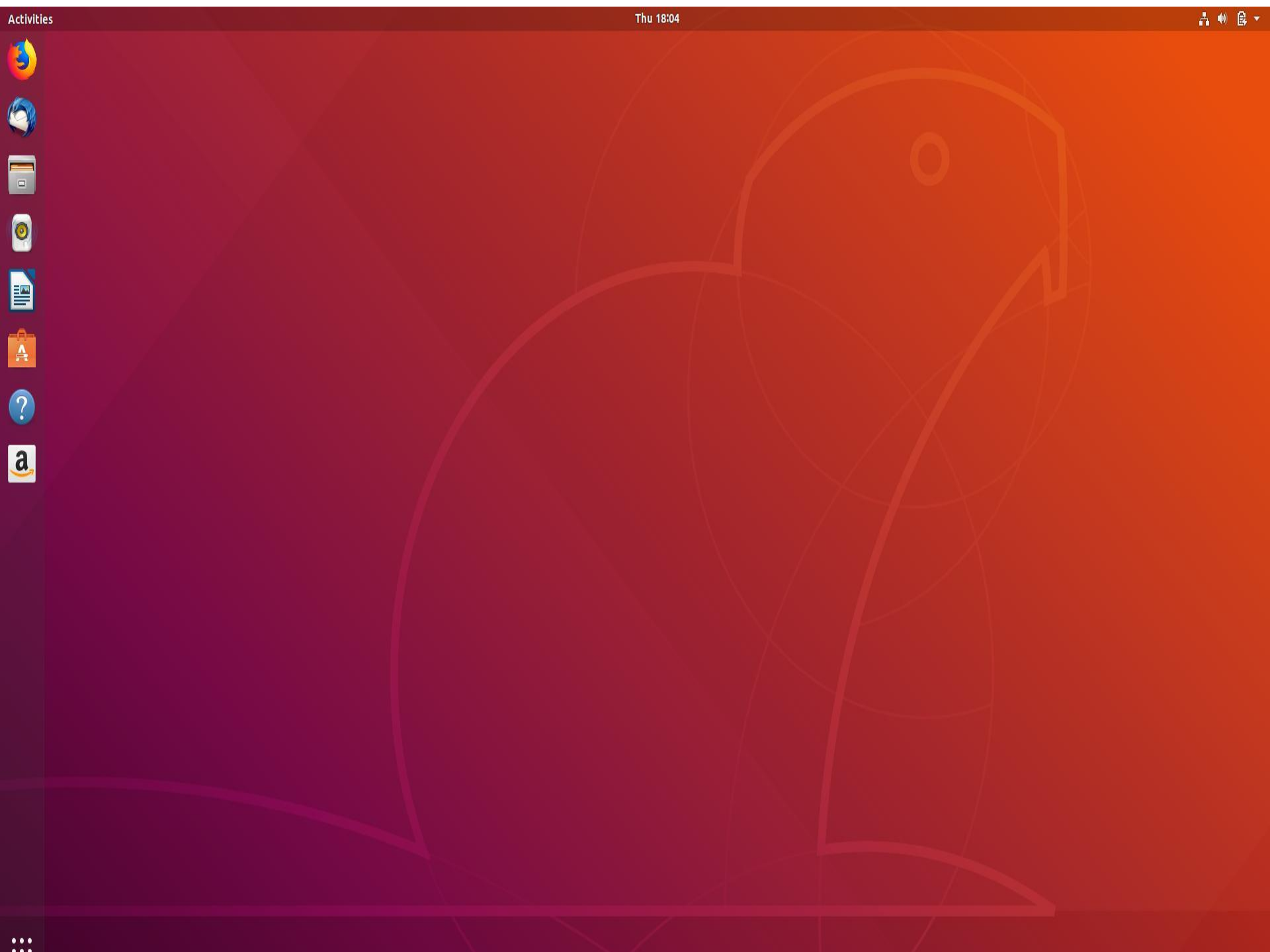
Code Improvements

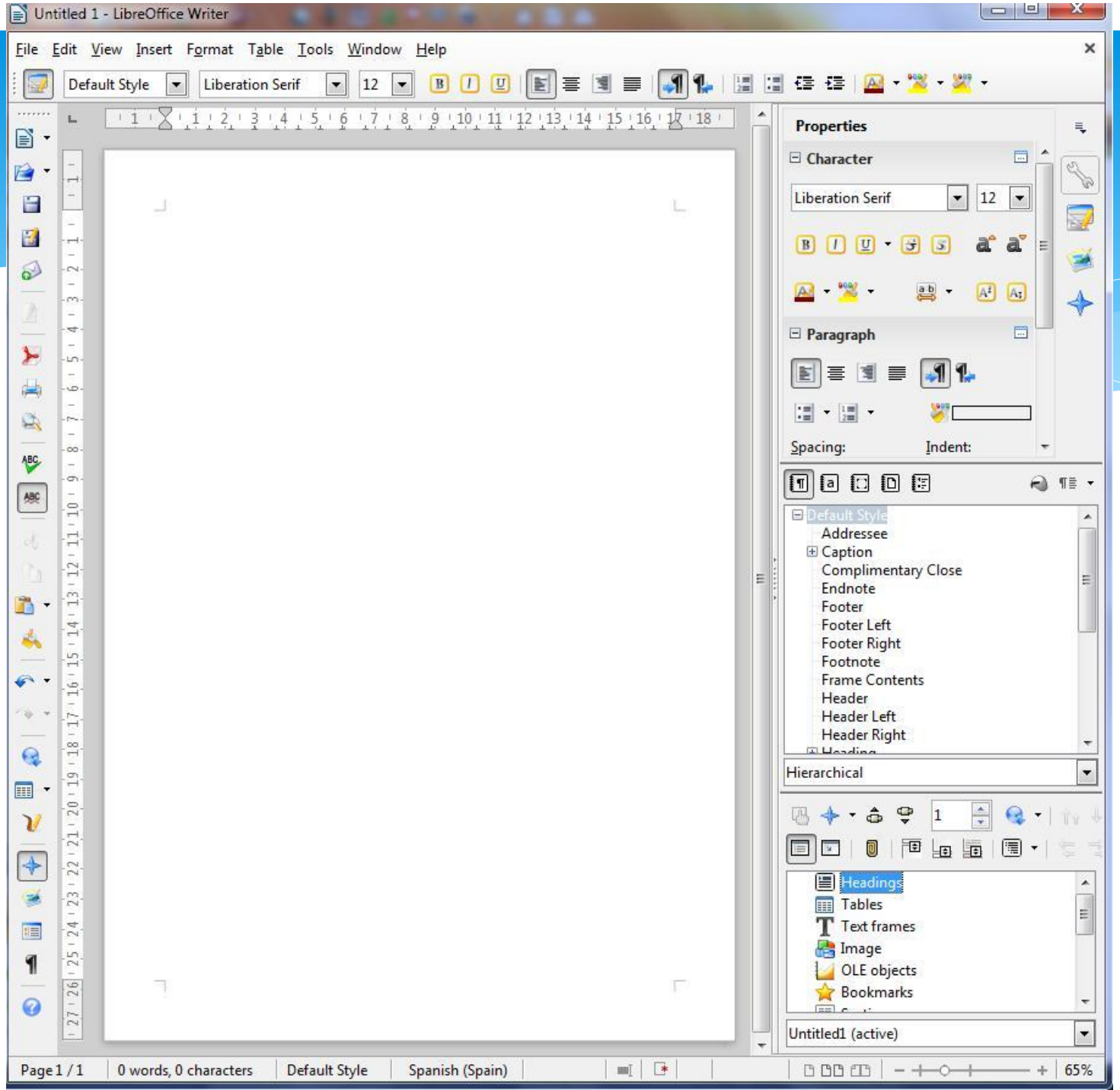
↓

New Release

Open-Source Product Examples

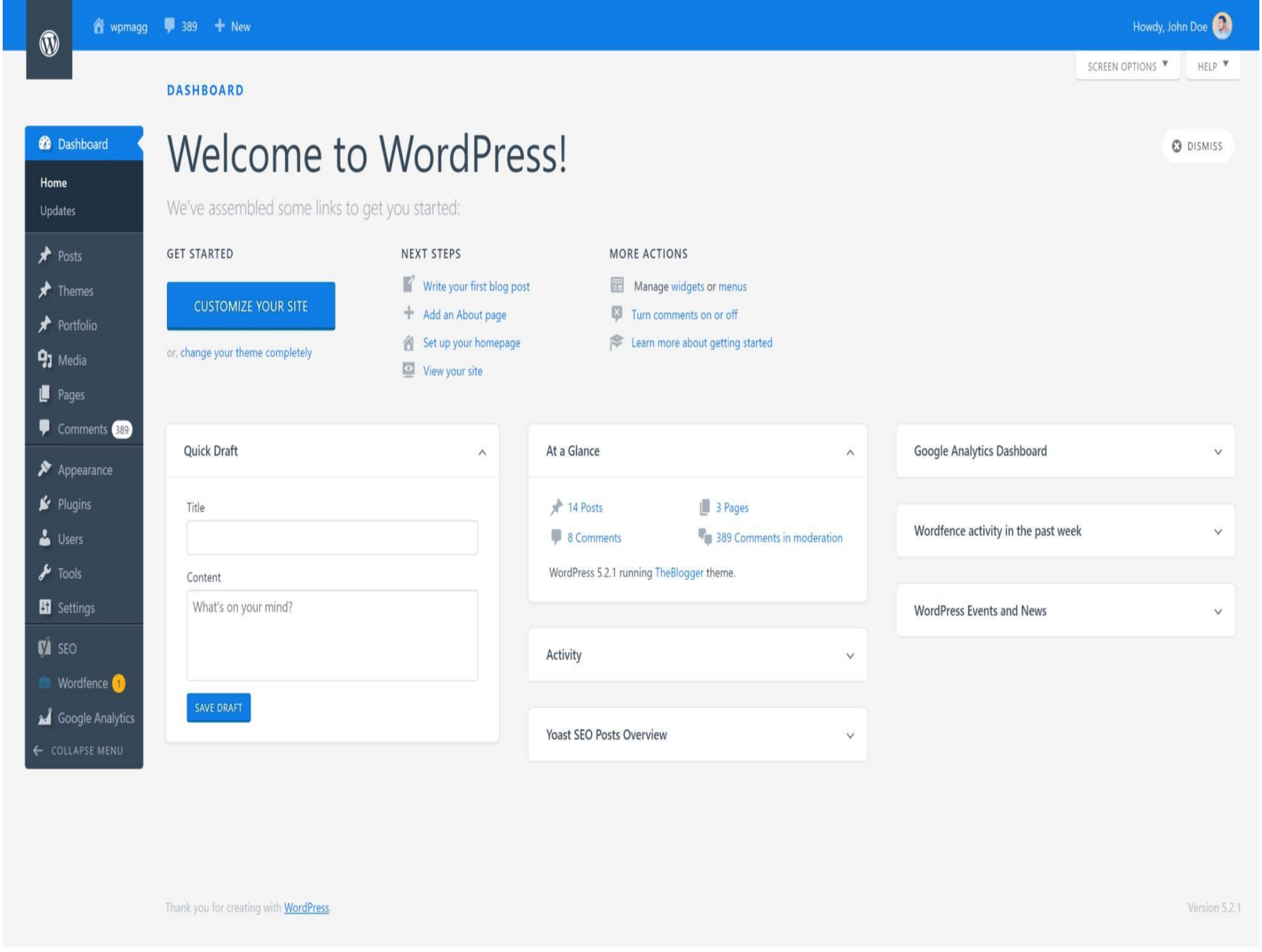
Product	Purpose
Ubuntu	Operating System
LibreOffice	Office Suite
Firefox	Web Browser
WordPress	Website Development
VLC	Multimedia
GIMP	Image Editing


Activities
Thu 18:04


Untitled 1 - LibreOffice Writer
File Edit View Insert Format Table Tools Window Help
Default Style
Liberation Serif
12
Properties
Character
Liberation Serif
12
Paragraph
Spacing:
Indent:
Default Style
Addressee
Caption
Complimentary Close
Endnote
Footer
Footer Left
Footer Right
Footnote
Frame Contents
Header
Header Left
Header Right
Hierarchical
Headings
Tables
Text frames
Image
OLE objects
Bookmarks
Untitled1 (active)
Page 1 / 1
0 words, 0 characters
Default Style
Spanish (Spain)
65%



Firefox®


wpmagg
389
New
Howdy, John Doe
SCREEN OPTIONS
HELP
DASHBOARD
Dashboard
Home
Updates
Posts
Themes
Portfolio
Media
Pages
Comments 389
Appearance
Plugins
Users
Tools
Settings
SEO
Wordfence 1
Google Analytics
COLLAPSE MENU
Welcome to WordPress!
DISMISS
We've assembled some links to get you started:
GET STARTED
CUSTOMIZE YOUR SITE
or, change your theme completely
NEXT STEPS
Write your first blog post
Add an About page
Set up your homepage
View your site
MORE ACTIONS
Manage widgets or menus
Turn comments on or off
Learn more about getting started
Quick Draft
Title
Content
What's on your mind?
SAVE DRAFT
At a Glance
14 Posts
3 Pages
8 Comments
389 Comments in moderation
WordPress 5.2.1 running TheBlogger theme.
Activity
Yoast SEO Posts Overview
Google Analytics Dashboard
Wordfence activity in the past week
WordPress Events and News
Thank you for creating with WordPress.
Version 5.2.1

Product	Purpose
Ubuntu	Operating System
LibreOffice	Office Suite
Firefox	Web Browser
WordPress	Website Development
VLC	Multimedia
GIMP	Image Editing

Feature Analysis Example

1. Ubuntu Linux:

* **Features**
  - Free operating system
  - High security
  - Stable
  - Large community
  - Frequent updates

* **Advantages**
  - No licensing cost
  - Supports programming
  - Customizable

* **Limitations**
  - Software compatibility issues
  - Learning curve for beginners

2. LibreOffice:

* **Features**
  - Word processing
  - Spreadsheet
  - Presentation
  - Database

* **Advantages**
  - Free
  - Compatible with Microsoft Office formats
  - Cross-platform

* **Limitations**
  - Some formatting differences

3. WordPress:

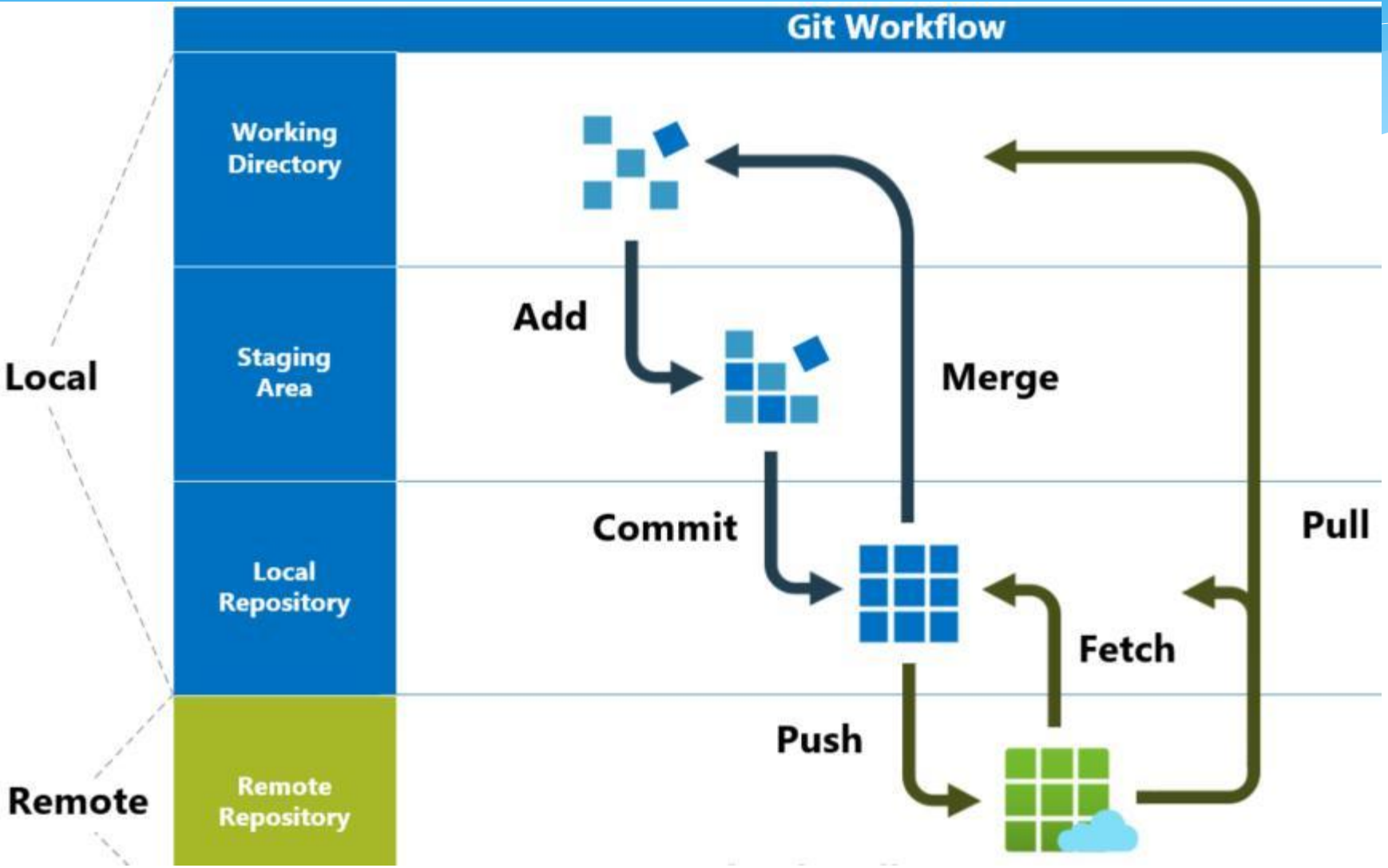
- **Features**
  - Website creation
  - Themes
  - Plugins
  - SEO support

- **Advantages**
  - Easy to learn
  - Large community
  - Thousands of plugins

- **Limitations**
  - Easy to learn
  - Large community
  - Thousands of plugins

- **Used by**
  - Universities
  - Government
  - Businesses

Git and GitHub in Open Source Development



MyFirstRepo

Public

Pin | Unwatch 1 | Fork 0 | Star 0

main | 2 branches | 0 tags

Go to file | Add file | Code

Your main branch isn't protected

Protect this branch from force pushing or deletion, or require status checks before merging. Learn more

Protect this branch | X

error505 Update README.md	68789b3 1 hour ago	6 commits
LICENSE	Initial commit	last month
README.md	Update README.md	1 hour ago
index.html	Update index.html	3 days ago
styles.css	Create styles.css	3 days ago

README.md

My First Repo

A brief description of your project goes here. Explain what it does and why it's useful.

Table of Contents

About

This is my first GitHub Repository

- Readme
- MIT license
- Activity
- 0 stars
- 1 watching
- 0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Languages



Basic Git Workflow

git init

1

Working Directory

Make changes to files:

- Additions
- Deletions
- Modifications

git add

2

Staging Area

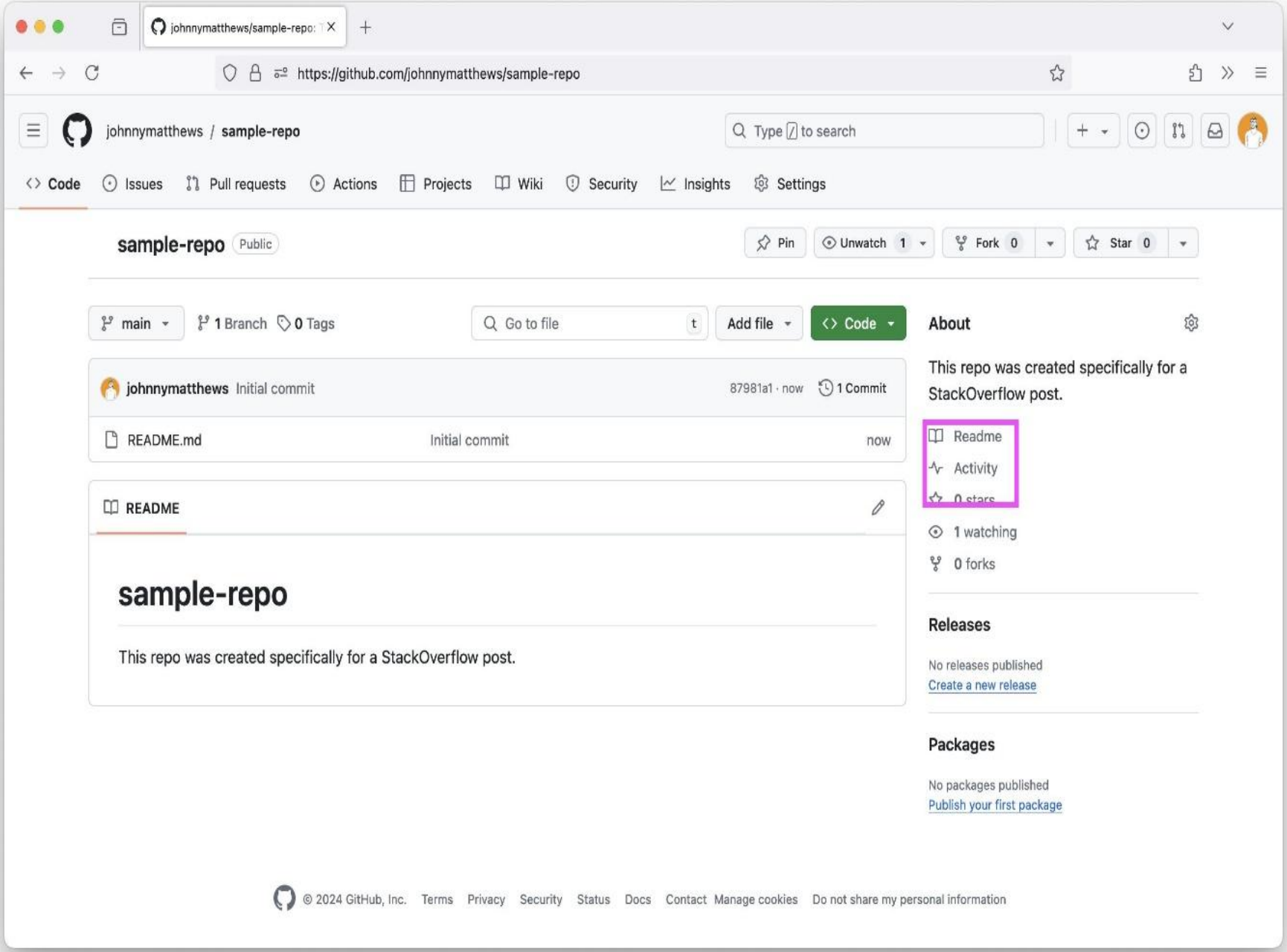
Files are ready for commit

git commit

3

Repository

Changes are saved to repository as a commit


johnnymatthews/sample-repo
https://github.com/johnnymatthews/sample-repo
johnnymatthews / sample-repo
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
sample-repo
Public
Pin
Unwatch 1
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
t
Add file
Code
johnnymatthews Initial commit
87981a1 · now
1 Commit
README.md
Initial commit
now
README
sample-repo
This repo was created specifically for a StackOverflow post.
About
This repo was created specifically for a StackOverflow post.
Readme
Activity
0 stars
1 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
© 2024 GitHub, Inc.
Terms
Privacy
Security
Status
Docs
Contact
Manage cookies
Do not share my personal information

- **Git**
  - Version Control System

- **GitHub**
  - Cloud repository
  - Collaboration
  - Issue tracking
  - Pull Requests
  - Code Review



* Git -
  - Version Control System

* GitHub -
  - Cloud repository
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  - Pull Requests
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Workflow

Clone

↓

Edit

↓

Commit

↓

Push

↓

Pull Request

↓

Merge

Benefits of Open Source Development

* Benefits
  - Cost reduction
  - Innovation
  - Knowledge sharing
  - Faster bug fixing
  - High reliability
  - Transparency
  - Better learning opportunities

Challenges of Open Source

* Limited official support
* Compatibility issues
* Documentation quality varies
* Requires technical knowledge
* Community participation needed

Open Source vs Proprietary Software

Open Source	**Proprietary**
Free	Paid
Source available	Source hidden
Community support	Vendor support
Customizable	Limited customization
Transparent	Closed development
Flexible	License restrictions

Real World Applications

Open Source is used in

* **Education -** Moodle
* **Healthcare -** OpenMRS
* **Government -** Linux servers
* **Business -** ERPNext
* **Cloud -** OpenStack
* **Artificial Intelligence –** TensorFlow, PyTorch

Open Source Development Process

Idea

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Planning

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Coding

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Git Repository

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Testing

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Community Review

↓

Documentation

↓

Release

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Maintenance

Case Study

Mozilla Firefox

* Features
  - Open source browser

* Benefits
  - Privacy
  - Extensions
  - Fast updates
  - Cross-platform

* Development
  - Thousands of contributors worldwide

Classroom Activity

Group Activity

* Select one OSS product.

* Analyze:
  - Features
  - Advantages
  - Disadvantages
  - Target users
  - Community support
  - Development process

* Present findings within 10 minutes.

THANK YOU